

Competitive Firms and Markets - SOLUTIONS

Practice Problems:

1. A consumer has a demand function for lemons of $q = 10 - 5p$. Find the elasticity of demand for lemons of this consumer, and comment on how it changes from $p = 0$ to $p = 2$.

$$\varepsilon = \frac{dq}{dp} \frac{p}{q} \Rightarrow \varepsilon = -\frac{5p}{q}$$

At $p = 0$, $\varepsilon = 0$ and so is perfectly inelastic, while at $p = 2$, $\varepsilon = \infty$ and so is perfectly elastic. Hence, elasticity of demand changes along the demand curve. Its gradient is then a kind of “average” across the entire curve.

2. Suppose that there are 5 identical firms in a competitive market such that the elasticity of demand is given by -0.5 . Suppose further that the elasticity of supply is 4.4 . Calculate the residual elasticity of demand for an individual firm.

The residual elasticity of demand is given by the equation:

$$\varepsilon_i = n\varepsilon - (n - 1)\eta_0$$

By substitution we obtain:

$$\varepsilon_i = (5 \times -0.5) - (5 - 1)4.4 = -2.5 - (4 \times 4.4) = -20.1$$

3. Consider a competitive market of 15 firms, where each firm has a supply function of $q_i = \frac{2}{5}p$, and where the elasticity of demand is -1.5 .
 - a. Find the market supply function and derive the elasticity of supply for $n - 1$ firms.

To find the market supply function we sum up all the individual ones:

$$Q = \sum_{i=1}^{15} q_i = \sum_{i=1}^{15} \frac{2}{5}p = 6p$$

However, for 14 firms it is given by:

$$Q_0 = \sum_{i=1}^{14} q_i = \sum_{i=1}^{14} \frac{2}{5}p = 5.6p$$

The elasticity of supply is:

$$\eta_0 = \frac{dQ_0}{dp} \frac{p}{Q_0} = \frac{5.6p}{Q_0}$$

- b. Use these values to derive the residual elasticity of demand for an individual firm.

We use equation given above:

$$\varepsilon_i = n\varepsilon - (n - 1)\eta_0$$

Substituting in all the values:

$$\varepsilon_i = (15 \times -1.5) - (15 - 1) \frac{5.6p}{Q_0}$$

$$\varepsilon_i = -22.5 - \frac{78.4p}{Q_0}$$

4. Prove that if a firm is maximising profits, then it produces at the output where its marginal revenue is equal to its marginal cost.

Profits are, by definition, given by:

$$\pi(q) = TR(q) - TC(q)$$

We take the first derivative with respect to quantity:

$$\frac{d\pi}{dq} = \frac{dTR}{dq} - \frac{dTC}{dq}$$

We are trying to find the stationary point, hence we set the derivative equal to zero:

$$\frac{d\pi}{dq} = \frac{dTR}{dq} - \frac{dTC}{dq} = 0$$

Rearranging yields the solution:

$$\frac{dTR}{dq} - \frac{dTC}{dq} = 0 \Rightarrow \frac{dTR}{dq} = \frac{dTC}{dq} \Rightarrow \mathbf{MR = MC}$$

5. For each of the following cases determine whether the firm will close down or continue operating. Assume in each case that firms are in the short run:
- $p = 5; q = 20; VC = 4q; FC = 150$

$$TR = 5 \times 20; VC = 4 \times 20 \Rightarrow \mathbf{TR = 100 > VC = 80}$$

OPERATE: Hence, the firm cannot reduce losses by stopping operations. This is because total revenue exceeds variable costs. Hence, the firm will continue to operate in the short run but will close down in the long run if a losses are not eliminated.

b. $TR = 125; VC = 126; FC = 1$

$$TR = 125 < VC = 126$$

SHUT DOWN: Since variable costs are greater than total revenue, a firm may reduce losses by stopping operations. Hence, it will close down in the short run.

c. $MR = p; TC = q^3; p = 3$

$$MR = p, MC = 3q^2 \Rightarrow 3q^2 = 3 \Rightarrow q = 1$$

At this quantity and price, it holds that:

$$\pi = 3 - 1 = 2 > 0$$

OPERATE: Hence the firm will continue operating and will even earn a profit! Notice that in this case there are no fixed costs.

d. $MR = p; TC = q^3 + 60; p = 27$

$$MR = p, MC = 3q^2 \Rightarrow 3q^2 = 27 \Rightarrow q = 3$$

At this quantity and price, it holds that:

$$\pi = 81 - (27 + 60) = -6 < 0$$

OPERATE: Even though profits are negative, we are dealing with the short run. If we compare revenue and variable costs ($81 - 27 > 0$), we will see that they are positive, and hence the firm will continue operating in the short-run but would shut down in the long run.

6. Consider the following situations and find the profit maximising level of output:

a. $MC = q^2 - 2q + 1; MR = q - 1$

$$q^2 - 2q + 1 = q - 1 \Rightarrow (q - 1)(q - 1) = (q - 1)$$

Since $q > 0$, this equation can be simplified to:

$$(q - 1) = 1 \Rightarrow q^* = 2$$

b. $\pi(q) = -q^2 + 4q + 10$

Here we are simply taking the first derivative and finding the stationary point by setting it equal to zero:

$$\frac{d\pi}{dq} = -2q + 4 = 0 \Rightarrow q^* = 2$$

7. A firm operating in a competitive market has two plants in which it can produce. The cost functions of each plant are: $C(q) = q^3 - 4q^2 + 2q$ and $C(q) = q^3 - 2q^2 + q$:

- a. Find the supply functions of each plant.

Since we are dealing in a competitive market, we have that:

$$p = 3q_1^2 - 8q + 2 \text{ and } p = 3q_2^2 - 4q + 1$$

Notice how this implies that the marginal costs are equal to each other anyway! We use the quadratic formula to find the supply curves of each plant:

$$3q_1^2 - 8q + (2 - p) = 0$$

$$q_1 = \frac{8 \pm \sqrt{64 - (4 \times 3 \times (2 - p))}}{6} = \frac{8 \pm \sqrt{40 + 12p}}{6}$$

We remember that everything must be positive here, hence:

$$q_1 = \frac{8 + \sqrt{40 + 12p}}{6}$$

The same procedure is taken for the entire plant:

$$3q_2^2 - 4q + (1 - p) = 0$$

$$q_2 = \frac{4 \pm \sqrt{16 - (4 \times 3 \times (1 - p))}}{6} = \frac{4 \pm \sqrt{4 + 12p}}{6}$$

We remember that everything must be positive here, hence:

$$q_2 = \frac{4 + \sqrt{4 + 12p}}{6}$$

- b. Derive the profit function of the entire firm.

The profit function is what arises when the firm is operating optimally. It expresses its optimal profit in terms of exogenous variables. In this way we have that:

$$\pi = p(q_1 + q_2) - c(q_1) - c(q_2)$$

We just substitute everything in:

$$\pi = p \left(\frac{8 + \sqrt{40 + 12p}}{6} + \frac{4 + \sqrt{4 + 12p}}{6} \right) - c(q_1) - c(q_2)$$

$$\pi = p \left(\frac{12 + \sqrt{40 + 12p} + \sqrt{4 + 12p}}{6} \right) - c(q_1) - c(q_2)$$

Now let's deal with the costs:

$$c(q_1) = q_1^3 - 4q_1^2 + 2q_1 = \left(\frac{8+\sqrt{40+12p}}{6}\right)^3 - 4\left(\frac{8+\sqrt{40+12p}}{6}\right)^2 + 2\left(\frac{8+\sqrt{40+12p}}{6}\right)$$

$$c(q_2) = q_2^3 - 2q_2^2 + q_2 = \left(\frac{4+\sqrt{4+12p}}{6}\right)^3 - 2\left(\frac{4+\sqrt{4+12p}}{6}\right)^2 + \left(\frac{4+\sqrt{4+12p}}{6}\right)$$

Hence the entire profit function is given by:

$$\begin{aligned} \pi = p \left(\frac{12 + \sqrt{40 + 12p} + \sqrt{4 + 12p}}{6} \right) &- \left(\frac{8 + \sqrt{40 + 12p}}{6} \right)^3 \\ &- 4 \left(\frac{8 + \sqrt{40 + 12p}}{6} \right)^2 + 2 \left(\frac{8 + \sqrt{40 + 12p}}{6} \right) \\ &- \left(\frac{4 + \sqrt{4 + 12p}}{6} \right)^3 - 2 \left(\frac{4 + \sqrt{4 + 12p}}{6} \right)^2 + \left(\frac{4 + \sqrt{4 + 12p}}{6} \right) \end{aligned}$$

You could probably simplify this, but I can't really be bothered to go through the entire process. If you got here, I think it's good enough! ☺

8. A competitive market operates in the short run with 4 firms. The market demand curve is given by: $Q = \frac{I+20p}{p}$, where I is aggregate consumer income. Suppose that the cost function of each firm is $C(q) = 5 + q^2$. Suppose that $I = 62.5$

- a. Find the individual and market supply curve.

We use the important condition for a PC market:

$$p = MC \Rightarrow p = 2q \Rightarrow q = 0.5p$$

This is the supply curve for just one firm. The market supply curve is given by the sum of the individual supply curves:

$$Q = 2p$$

- b. What will be the equilibrium price?

Set the market demand and supply curve:

$$\frac{62.5 + 20p}{p} = 2p \Rightarrow 2p^2 - 20p - 62.5 = 0 \Rightarrow p > 0, p = 12.5$$

- c. How much profit will each firm make?

Each firm is identical, and their profit equation is given by:

$$\pi = qp - 5 - q^2 = 0.5p^2 - 5 - 0.25p^2 = 0.25p^2 - 5 = 34.0625$$

- d. How do these change when consumer income falls to $I = 10.5$?

Set the market demand and supply curve:

$$\frac{10.5 + 20p}{p} = 2p \Rightarrow 2p^2 - 20p - 10.5 = 0 \Rightarrow p > 0, p = \mathbf{10.5}$$

$$\pi = pq - 5 - q^2 = 0.5p^2 - 5 - 0.25p^2 = 0.25p^2 - 5 = \mathbf{22.5625}$$

As demand decreases, the profits of the firms decline.

9. Suppose now that there is a competitive market operating in the long run with the demand function $Q = \frac{100+20p}{p}$. As in the previous question each firm that either operates in the market is looking to enter has the cost function $C(q) = 5 + q^2$. Use what you know about competitive markets to work out the equilibrium number of firms.

Use that price equals marginal cost:

$$p = 2q \Rightarrow q = 0.5p$$

Market supply depends on the number of firms, so does the equilibrium price:

$$Q = 0.5np \Rightarrow \frac{100 + 20p}{p} = 0.5np$$

$$100 + 20p = 0.5np^2$$

$$0.5np^2 - 20p - 100 = 0$$

Using the quadratic formula, we have that:

$$p = \frac{20 \pm \sqrt{400 - (4 \times 0.5n \times -100)}}{n} = \frac{20 \pm \sqrt{400 - (4 \times 0.5n \times -100)}}{n}$$

$$p = \frac{20 + \sqrt{400 + 200n}}{n}, p > 0$$

The profit function of each firm is then:

$$\pi = pq - c(q) = p(0.5p) - 5 - 0.25p^2 = 0.25p^2 - 5 = \frac{20 + \sqrt{400 + 200n}}{4n} - 5$$

We know that in the long run, profit is zero. Hence, we have that:

$$\frac{20 + \sqrt{400 + 200n}}{4n} = 5 \Rightarrow 400 + 200n = (4n - 20)^2$$

$$16n^2 - 160n + 400 = 400 + 200n \Rightarrow 16n^2 - 360n = 0 \Rightarrow n > 0, n = \mathbf{22.5}$$

Hence, the number of firms in the long run in this market will be 22.5. Of course, keep in mind that we are technically in violation of some subtle assumptions on competitive markets: namely that the number of firms is infinite or tends to infinity. We have set this idea to the side for mathematical tractability.

10. A firm operates with the production function $q = K^{0.5}L^{0.5}$. Assuming that inputs prices are $w = r = 1$.
- Finds its expansion path and cost function.

Set MRTS equal to slope of isocost line yielding:

$$K = L$$

This should be readily apparent because the ratio of input prices is one, and the exponents on the Cobb-Douglas production function are equal as well. Then, we look at total cost:

$$TC = K + L = 2L$$

Now we need to find labour in terms of output. Substitute expansion path into production function:

$$q = L^{0.5}L^{0.5} = L$$

Hence, the cost function is simply:

$$TC = 2q$$

- From now on, suppose there are 19 other firms just like this one operating in a competitive market with demand function $Q = 100 - P$. Find the individual supply curve for each firm:

This is a **TRICK** question. Let's formulate:

$$\pi = pq - 2q = q(p - 2)$$

If one looks at this profit function, you should see that if price is greater than 2, firms can earn infinite profits by increasing output. If price is less than 2, then the firm produces nothing, while if price is 2, then it can produce anything it wants and will earn 0 profits. Hence the supply curve for an individual firm is:

$$S = \begin{cases} q = \infty, p > 2 \\ q = [0, \infty), p = 2 \\ q = 0, p < 2 \end{cases}$$

11. **(Challenge)** By using the residual demand curve, prove that the individual elasticity of demand for a firm is given by:

$$\varepsilon_i = n\varepsilon - (n - 1)\eta_0$$

We use the residual demand curve:

$$D_i^r(p) = D(p) - S_0(p)$$

$$\frac{\partial D_i^r}{\partial p} = \frac{\partial D}{\partial p} - \frac{\partial S_0}{\partial p} \Rightarrow \frac{\partial D_i^r}{\partial p} \frac{p}{q_i} = \frac{\partial D}{\partial p} \frac{p}{Q} \frac{Q}{q_i} - \frac{\partial S_0}{\partial p} \frac{p}{Q_0} \frac{Q_0}{q_i}$$

Note the following facts:

$$\varepsilon_i = \frac{\partial D_i^r}{\partial p} \frac{p}{q_i}; \varepsilon = \frac{\partial D}{\partial p} \frac{p}{Q}; n = \frac{Q}{q_i}; \eta_0 = \frac{\partial S_0}{\partial p} \frac{p}{Q_0}; n - 1 = \frac{Q_0}{q_i}$$

By substitution, we get the desired result:

$$\varepsilon_i = n\varepsilon - (n - 1)\eta_0$$

12. **(Challenge)** Suppose that we are operating in the short run in a competitive market. Suppose all firms are identical and have a Cobb-Douglas production function of the form $q_i = K^\alpha L^\beta$. Suppose further that input prices are $w = 1$; $r = 1$.

- a. Use the production function and given input prices to derive the expansion path of each firm.

$$\frac{\beta K}{\alpha L} = 1 \Rightarrow K = \frac{\alpha}{\beta} L$$

- b. Use the expansion path to derive the cost function for each firm.

$$q = \left(\frac{\alpha}{\beta} L\right)^\alpha L^\beta = \left(\frac{\alpha}{\beta}\right)^\alpha L^{(\alpha+\beta)}$$

$$L = \left(q \left(\frac{\alpha}{\beta}\right)^{-\alpha}\right)^{\frac{1}{\alpha+\beta}}$$

$$C(q) = L + K = L + \frac{\alpha}{\beta} L = \left(1 + \frac{\alpha}{\beta}\right) \left(q \left(\frac{\alpha}{\beta}\right)^{-\alpha}\right)^{\frac{1}{\alpha+\beta}}$$

Hence, the cost function is:

$$C(q) = \left(1 + \frac{\alpha}{\beta}\right) \left(\frac{\alpha}{\beta}\right)^{\frac{-\alpha}{\alpha+\beta}} q^{\frac{1}{\alpha+\beta}}$$

- c. State the profit maximisation problem.

$$\max_q \pi = pq - \left(1 + \frac{\alpha}{\beta}\right) \left(\frac{\alpha}{\beta}\right)^{\frac{-\alpha}{\alpha+\beta}} q^{\frac{1}{\alpha+\beta}}$$

- d. Use the fact that the firms are operating in a competitive market to derive the individual supply curve of each firm.

Use the fact that price equals marginal cost:

$$p = \frac{1}{\alpha + \beta} \left(1 + \frac{\alpha}{\beta}\right) \left(\frac{\alpha}{\beta}\right)^{-\frac{\alpha}{\alpha+\beta}} q^{\frac{1-\alpha-\beta}{\alpha+\beta}}$$

Rearranging yields:

$$q = \left(\frac{\alpha + \beta}{1 + \frac{\alpha}{\beta}} \left(\frac{\alpha}{\beta}\right)^{\frac{\alpha}{\alpha+\beta}} p \right)^{-\frac{\alpha+\beta}{1-\alpha-\beta}}$$

- e. Use the individual supply curve, derive the profit function of each firm.

Substituting into the profit maximisation equation yields:

$$\begin{aligned} \pi = & p \left(\frac{\alpha + \beta}{1 + \frac{\alpha}{\beta}} \left(\frac{\alpha}{\beta}\right)^{\frac{\alpha}{\alpha+\beta}} p \right)^{-\frac{\alpha+\beta}{1-\alpha-\beta}} \\ & - \left(1 + \frac{\alpha}{\beta}\right) \left(\frac{\alpha}{\beta}\right)^{\frac{-\alpha}{\alpha+\beta}} \left(\left(\frac{\alpha + \beta}{1 + \frac{\alpha}{\beta}} \left(\frac{\alpha}{\beta}\right)^{\frac{\alpha}{\alpha+\beta}} p \right)^{-\frac{\alpha+\beta}{1-\alpha-\beta}} \right)^{\frac{1}{\alpha+\beta}} \end{aligned}$$

Simplifying yields the profit function:

$$\begin{aligned} \pi = & p^{\frac{1-2\alpha-2\beta}{1-\alpha-\beta}} \left(\frac{\alpha + \beta}{1 + \frac{\alpha}{\beta}} \right)^{-\frac{\alpha+\beta}{1-\alpha-\beta}} \left(\frac{\alpha}{\beta}\right)^{-\frac{\alpha}{1-\alpha-\beta}} \\ & - \left(1 + \frac{\alpha}{\beta}\right) \left(\frac{\alpha}{\beta}\right)^{\frac{\alpha-\alpha(1-\alpha-\beta)}{(\alpha+\beta)(1-\alpha-\beta)}} \left(\frac{\alpha + \beta}{1 + \frac{\alpha}{\beta}} \right)^{-\frac{1}{1-\alpha-\beta}} p^{-\frac{1}{1-\alpha-\beta}} \end{aligned}$$